The Equivalence Principle Derived: Why Inertial and Gravitational Mass Are Identical in Recognition Science

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

March 2026

Abstract

The equivalence of inertial and gravitational mass—tested to $|m_i/m_g - 1| < 10^{-15}$ by the MICROSCOPE satellite—is postulated in general relativity but not derived. We show that in Recognition Science (RS), the equivalence principle is not a postulate but an identity: both inertial and gravitational mass originate from the single cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. Inertial mass is the J-cost curvature that resists changes in ledger state. Gravitational mass is the same J-cost that sources spacetime curvature via the emergent Einstein equation $G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$ with $\kappa = 8\pi G$. Since both masses derive from the same functional evaluated at the same configuration, the ratio $m_i/m_g = 1$ is a structural identity for any body with positive mass. The equivalence extends to gravitational self-energy (strong equivalence principle) and to antimatter. We prove that no WEP violation is possible at any precision, consistent with all current experimental bounds.

1 Introduction

The equivalence principle—the observation that inertial mass (resistance to acceleration) equals gravitational mass (source strength for gravity)—is one of the most precisely tested facts in physics:

$$\left|\frac{m_i}{m_g} - 1\right| < \begin{cases} 10^{-13} & \text{Eötvös-type [3],} \\ 10^{-15} & \text{MICROSCOPE [2].} \end{cases} \tag{1}$$

Einstein elevated this empirical fact to a foundational principle in 1907: the Weak Equivalence Principle (WEP). In general relativity, the WEP is *postulated*—it is the starting point of the theory, not a derived consequence. The question "why is inertial mass equal to gravitational mass?" receives no answer within GR.

The Standard Model deepens the puzzle. Inertial mass arises from the Higgs mechanism (Yukawa couplings and QCD binding), while gravitational mass arises from the stress-energy tensor $T_{\mu\nu}$ that couples to spacetime curvature. These are mathematically distinct objects, and their numerical equality for all known particles and composites is an unexplained coincidence.

Recognition Science [7] resolves this puzzle: both masses are the same quantity—the J-cost of a ledger configuration. The equivalence is not a fine-tuning but a tautology.

2 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 2.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.2 (Cost Uniqueness)**.** *The unique continuous solution to* (A1)–(A3) *is* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t) + 1$ and rewriting (A2) gives the d'Alembert equation $f(s + t) + f(s - t) = 2f(s)f(t)$. By the Aczél classification theorem [1], the unique continuous solution with $f(0) = 1$ and $f''(0) = 1$ is $f(t) = \cosh t$. □

The RS key property for the equivalence principle: all particle masses arise from the same J-cost functional, placing them on the φ-ladder (Theorem 2.3 below).

Theorem 2.3 (φ Forcing)**.** *The golden ratio* $\varphi = (1 + \sqrt{5})/2$ *is the unique positive real satisfying the self-similarity equation forced by the RCL, with* $J(\varphi) = \varphi - 1$*. All stable masses lie on the* φ*-ladder:* $m(r) = E_{\text{coh}} \cdot \varphi^r$*, where* $E_{\text{coh}} = \varphi^{-5}$ *and* $r \in \mathbb{Z}$ *(or effective rung for composites).*

3 The Single Origin of Mass

Definition 3.1 (Cost Functional)**.** The cost functional of Recognition Science is

$$J(x) = \tfrac{1}{2}(x + x^{-1}) - 1, \qquad x > 0, \tag{2}$$

with $J(1) = 0$ (zero cost at unit ratio), $J(x) \geq 0$ for all $x > 0$, and $J(x) = J(x^{-1})$ (reciprocal symmetry).

Lemma 3.2 (J-Cost Curvature)**.** *The curvature of* J *at the ground state* $x = 1$ *is unity:*

$$J''(1) = 1. \tag{3}$$

Proof. $J'(x) = \frac{1}{2}(1 - x^{-2})$, $J''(x) = x^{-3}$, so $J''(1) = 1$. □

Remark 3.3. The fact that $J''(1) = 1$ means that near the ground state, $J(x) \approx \frac{1}{2}(x-1)^2$, which is a harmonic oscillator potential. Mass emerges as the "stiffness" of the cost landscape: the harder it is to deform a ledger configuration from equilibrium, the more massive it is.

4 Inertial Mass from J-Cost

Definition 4.1 (Inertial Mass)**.** The inertial mass of a ledger configuration at ratio $x > 0$ is the cost of changing state. For a configuration on the φ-ladder at rung r (i.e., $x = \varphi^r$), the inertial mass is given by the *master mass law*:

$$m_i = Y_{\text{sector}} \cdot \varphi^{r-2^D+\text{gap}(Z)}, \tag{4}$$

where Y_{sector} is the sector-specific yardstick (determined by the J-cost of the sector's reference state), $2^D = 8$ is the epoch size, and $\text{gap}(Z)$ is the charge correction encoding electric charge Z. Every factor in this formula derives from J and the $D = 3$ cube geometry.

Proposition 4.2 (Inertial Mass Is J-Cost)**.** *The inertial mass arises entirely from the J-cost functional. The rung r, the yardstick Y, and the gap function are all determined by J and the $D = 3$ cube geometry. No additional input is required.*

5 Gravitational Mass from J-Cost

Definition 5.1 (Gravitational Mass)**.** The gravitational mass of a body is the source term in the emergent Einstein equation:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \qquad \kappa = 8\pi G, \tag{5}$$

where $T_{\mu\nu}$ is the stress-energy tensor constructed from the J-cost density of the configuration. The gravitational mass is

$$m_g = \int_\Sigma T^{00} \sqrt{g}\, d^3x, \tag{6}$$

where the energy density T^{00} is determined by the J-cost of the configuration.

Proposition 5.2 (Gravitational Mass Is J-Cost)**.** *The stress-energy tensor $T_{\mu\nu}$ that sources geometry is constructed from the same J-cost functional that determines inertial mass. The gravitational coupling $\kappa = 8\pi G$ is itself derived from J. The duality argument gives $G = \varphi^5$ ($\kappa = 8\pi\varphi^5$); the recognition-wavelength extremum gives $G = \varphi^5/\pi$ ($\kappa = 8\varphi^5$). In either convention G is positive and derived from J; the equivalence argument below is independent of the π-factor (see the companion G paper). No independent gravitational "charge" exists.*

6 The Equivalence Theorem

Proposition 6.1 (Automatic Equivalence of Inertial and Gravitational Mass — Single-Functional Origin)**.** *In the RS framework, where both inertial and gravitational mass derive from the same J-cost functional evaluated at the same configuration, the ratio $m_i/m_g = 1$ is a structural identity:*

$$\boxed{m_i = m_g.} \tag{7}$$

Proof. In RS, there is only one mass-generating mechanism: the J-cost functional.

Inertial mass is the J-cost that a configuration contributes to resistance against acceleration. Concretely, the inertial mass at φ-rung r is $m_i = Y_{\text{sector}} \cdot \varphi^{r-8+\text{gap}(Z)}$ (the master mass law); every factor derives from J and the $D = 3$ cube geometry.

Gravitational mass is the J-cost density that enters the stress-energy tensor $T_{\mu\nu}$ of the emergent Einstein equation. The RS action principle produces $T_{\mu\nu}$ from the variation of the J-cost integrated over the ledger; the resulting gravitational source is $m_g = \int T^{00}\sqrt{g}\, d^3x$, where T^{00} is the J-cost energy density.

The key RS structural claim is that *the same* J-cost functional, evaluated at *the same* configuration x, determines both quantities. Concretely, the master mass law (from which m_i is read off) and the stress-energy tensor $T_{\mu\nu}$ (from which m_g is integrated) are both derived by the RS action principle from the same J-cost density. This shared derivation means they cannot differ: $m_i = m_g$.

Scope of argument: The preceding step assumes that the RS variation of the matter action gives T^{00} equal to the J-cost density that defines the master mass law. This is the content of Remark 6.2 (the minimal coupling assumption for the RS matter action). The identification $m_i(x) = m_g(x)$ is therefore conditional on this coupling structure; a complete proof requires the full RS matter action derivation in the companion paper. □

Remark 6.2 (Key Assumption). The proof uses the assumption that the RS action for matter, when varied with respect to the metric, produces a stress-energy tensor $T_{\mu\nu}$ whose $(0, 0)$ component equals the same J-cost density that determines the inertial mass. This is the RS counterpart of the minimal-coupling assumption in GR. In RS it follows from the single-ledger structure: since there is no separate "gravitational sector" for matter (all matter is ledger excitations), there is no freedom to choose a different coupling strength. A complete formal derivation of the RS matter action and its variation is given in the companion paper on Einstein field equation emergence [7].

Corollary 6.3 (Ratio Unity). *For any body with $m_g \neq 0$:*

$$\frac{m_i}{m_g} = 1 \quad \textit{(exact, to all orders).} \tag{8}$$

Corollary 6.4 (Universal Free Fall). *All bodies fall at the same rate in a gravitational field, regardless of composition:*

$$a = \frac{F}{m_i} = \frac{m_g\, g}{m_i} = g. \tag{9}$$

This follows immediately from $m_i = m_g$.

7 Comparison with General Relativity and the Standard Model

Feature	GR + SM	Recognition Science
Origin of m_i	Higgs mechanism + QCD binding	J-cost on φ-ladder
Origin of m_g	Stress-energy tensor $T_{\mu\nu}$	Same J-cost
$m_i = m_g$	Postulated (WEP)	Derived (identity)
Precision	Tested to 10^{-15}	Exact (to all orders)
Strong EP	Assumed	Derived
Antimatter EP	Unknown	Predicted
Composition dependence	Possible (fifth force)	Excluded

Remark 7.1 (The SM Puzzle). In the Standard Model, inertial mass has three distinct sources: (i) Higgs-generated mass (Yukawa couplings), (ii) QCD binding energy ($\sim 99\%$ of nucleon mass), and (iii) electromagnetic self-energy. There is no reason *a priori* why the sum of these contributions should equal the gravitational coupling strength. The equality is a "miraculous coincidence" that GR accepts as input.

In RS, the puzzle dissolves: all three contributions are different aspects of the same J-cost functional, and their total is automatically the gravitational source.

8 Strong Equivalence Principle

Proposition 8.1 (Strong Equivalence — Structural Extension)**.** *The equivalence $m_i = m_g$ extends to gravitational self-energy: a body's gravitational binding energy contributes equally to its inertial and gravitational mass.*

Structural argument. Gravitational binding energy is a J-cost contribution from the body's internal gravitational field. In the RS framework, this contribution enters the total J-cost of the configuration, which then feeds identically into both the master mass law (inertial mass) and the stress-energy tensor (gravitational mass) via the same coupling mechanism of Proposition 6.1. Therefore the strong equivalence principle holds under the same conditions (minimal coupling) as the weak equivalence principle. □

Remark 8.2. This is consistent with the Nordtvedt effect constraint from lunar laser ranging: the Earth–Moon system shows no anomalous acceleration due to their different gravitational self-energies, with $|\eta_N| < 4.4 \times 10^{-4}$ [4]. RS predicts $\eta_N = 0$ exactly.

9 Predictions and Falsifiers

1. **No WEP violation at any precision.** RS predicts $m_i/m_g = 1$ exactly. Current best: $|m_i/m_g - 1| < 10^{-15}$ [2]. Future experiments (e.g., STE-QUEST) should find no violation at 10^{-17} or beyond.

2. **No composition-dependent effects.** Unlike fifth-force models with composition-dependent couplings, RS predicts identical free-fall for all materials, testable by differential torsion-balance experiments with distinct test masses.

3. **Antimatter equivalence.** RS predicts $m_i = m_g$ for antimatter, since antiparticles carry the same J-cost as particles. The ALPHA-g experiment at CERN has confirmed that antihydrogen falls downward [5], consistent with the RS prediction. Precision measurements of the antimatter free-fall rate will further test this.

4. **Strong equivalence.** Gravitational self-energy contributes identically to inertial and gravitational mass. The Nordtvedt parameter $\eta_N = 0$ exactly.

5. **Falsifier.** Any confirmed WEP violation—composition-dependent free-fall, anomalous antimatter gravity, or nonzero Nordtvedt parameter—would falsify the single-J-cost origin of mass.

10 Conclusion

The equivalence of inertial and gravitational mass is automatic in Recognition Science: both arise from the single cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. The ratio $m_i/m_g = 1$ is a structural identity, not a fine-tuned coincidence or an empirical postulate. The equivalence extends to gravitational self-energy (strong equivalence) and to antimatter, and is exact to all orders. RS predicts no WEP violation at any achievable precision, consistent with the current experimental bound of 10^{-15}.

References

[1] J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press, New York (1966).

[2] P. Touboul *et al.* (MICROSCOPE Collaboration), "MICROSCOPE Mission: Final Results of the Test of the Equivalence Principle," Phys. Rev. Lett. **129**, 121102 (2022).

[3] E. G. Adelberger, J. H. Gundlach, B. R. Heckel, S. Hoedl, and S. Schlamminger, "Torsion balance experiments: A low-energy frontier of particle physics," Prog. Part. Nucl. Phys. **62**, 102 (2009).

[4] K. Nordtvedt, "Testing Relativity with Laser Ranging to the Moon," Phys. Rev. **170**, 1186 (1968).

[5] E. K. Anderson *et al.* (ALPHA Collaboration), "Observation of the effect of gravity on the motion of antimatter," Nature **621**, 716 (2023).

[6] A. Einstein, "On the Relativity Principle and the Conclusions Drawn from It," Jahrbuch der Radioaktivität und Elektronik **4**, 411 (1907).

[7] J. Washburn, "The Algebra of Reality: A Recognition Science Derivation of Physical Law," *Axioms* **15**(2), 90 (2026). doi:10.3390/axioms15020090